

BiofilmQ-HT

"Harnessing Bacteriophages for Human Welfare"

About This Utility

BiofilmQ-HT is a specialized, high-throughput software tool designed for researchers performing Crystal Violet biofilm assays. It computes the Specific Biofilm Formation (SBF) Index and percentage inhibition by directly processing raw Excel exports from Tecan i-control microplate readers.

This utility automates complex plate map assignments, performs robust blank subtractions across multiple media conditions, and automatically classifies biofilm strength. It is built to save hours of manual calculation while significantly reducing human error and improving reproducibility.

1. The 3-Step Workflow

The utility is structured into three intuitive steps to streamline the journey from raw data to publication-ready results.

Step 1 Design your Plate Map

Map the physical layout of your 96-well plate before analysis. This designates the sample types in each well.

- **Well Types:** Assign wells as *Strain/Condition*, *Blank*, *Positive Ctrl*, *Negative Ctrl*, or *Empty*.
- **Selection Tools:** Click individual wells or use the "**Select region**" tool to drag and highlight blocks of wells for bulk assignment.
- **Media Conditions (Crucial):** Specify the growth media for each well (e.g., "Default", "LB 100%"). The software automatically pairs Strains/Conditions with Blanks that share the **exact same Media Condition name** for accurate background subtraction.
- **Saving Templates:** Export your layout as a JSON template to instantly reload it for future experiments.

Step 2 Upload Data & Set Thresholds

Upload two separate Excel (.xlsx) files exported directly from your Tecan reader:

1. **OD 600 nm:** Growth measurement (cell density/turbidity).
2. **OD 570 nm:** Crystal Violet stain measurement (biofilm biomass).

Formatting Note: Do not modify the Tecan Excel files before uploading. The software relies on the raw <> anchor tags generated by the Tecan i-control software to locate the grid.

Formulas & Thresholds:

$$\text{SBF Index} = (\text{OD}_{570 \text{ sample}} - \text{OD}_{570 \text{ blank}}) / (\text{OD}_{600 \text{ sample}} - \text{OD}_{600 \text{ blank}})$$

$$\% \text{ Inhibition} = 100 - [(OD_{570} \text{ treated} - \text{blank}) / (OD_{570} \text{ pos ctrl} - \text{blank}) \times 100]$$

Adjust classification thresholds as needed. Default values are:

- **Weak Biofilm** SBF ≥ 0.5
- **Moderate Biofilm** SBF ≥ 1.0
- **Strong Biofilm** SBF ≥ 2.0

Step 3 Analyze Results & Export

Click "Calculate SBF & Inhibition" to generate your dashboard:

- **Summary Dashboard:** High-level overview of total strains/conditions, media types, and biofilm strength distribution.
- **SBF Bar Chart:** Visual ranking of all strains/conditions by their SBF Index.
- **Detailed Data Table:** Displays mean ODs, blank values, standard deviations, SBF index, and % Inhibition for every replicate group.
- **Plate Heatmap:** A visual 96-well plate representation, colored by final biofilm classification.

Export your final data as a CSV (for Prism/Excel), JPEG (for presentations), or PDF (for lab notebooks).

2. Common Experimental Workflows

Workflow A: Standard Biofilm Screening

Goal: Screen a library of isolates or environmental conditions to determine baseline biofilm capabilities.

Configuration Step	Instruction
Plate Map	Assign samples as <i>Strain/Condition</i> wells. Leave Media as Default. Assign at least one set of <i>Blank</i> wells (also Default media).
Analysis Focus	Evaluate the SBF Index column and the categorical tags (Strong , Moderate , etc.).

Workflow B: Biofilm Inhibition (Antibiotics/Phages)

Goal: Test how experimental treatments inhibit biofilm formation compared to an untreated control.

Configuration Step	Instruction
Positive Controls	You must assign wells as <i>Positive ctrl</i> to represent your untreated growth reference.
Strain/Condition	Assign treated samples as <i>Strain/Condition</i> wells. Name them by treatment (e.g., "Amp300 + Phage 1").
Analysis Focus	Evaluate the % Inhibition column. The software automatically compares the treated well's corrected OD570 against the Positive Ctrl's corrected OD570.

Workflow C: Media Optimization & Multi-Condition

Goal: Test the same strain/condition across multiple media types (e.g., 100% LB vs. 25% LB) simultaneously.

Configuration Step	Instruction
Media Matching	Type the specific media name in the <i>Media / Condition</i> field (e.g., 100% LB) during plate mapping.
Blank Assignment	Assign separate <i>Blank</i> wells for each media condition used (e.g., Blanks labeled 100% LB and 25% LB).
Smart Subtraction	The tool groups data by Media Condition, automatically subtracting the 100% LB blank average from only the 100% LB samples.